Geospatial Analysis of COVID-19 Vaccination, A Neighbourhood Level Analysis in Ireland

A Preprint

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Abstract

Despite Ireland's high national COVID-19 vaccination coverage, substantial disparities remain at the neighbhourhood levels. This thesis presents a comprehensive neighbourhood-level geospatial analysis of vaccine uptake across Ireland's 166 Local Electoral Areas (LEAs), integrating demographic, spatial, and health service accessibility data to identify structural and socio-demographic determinants in peoplw 12 years and older. Using an ecological design and publicly available data, we constructed a unified analytical dataset comprising vaccine uptake rates, population composition, deprivation metrics, and travel-time-based accessibility scores.A suite of Bayesian regression models—including spatial and hierarchical temporal components—was implemented to evaluate predictors of primary vaccine dose uptake. Initial models revealed that age (particularly 55–70 years) and upper secondary educational attainment were consistent, strong positive predictors, while health status and physical access to vaccination centres showed negligible influence. Spatial autocorrelation detected via Moran's I justified the use of Conditional Autoregressive (CAR) models, which confirmed neighbourhood-level clustering not explained by covariates. A four-parameter logistic growth model further captured LEA-specific vaccination trajectories over time, showing that while uptake timing was relatively synchronized across regions, final coverage levels varied significantly by local demographics.Contrary to expectations, broader deprivation indices did not significantly improve model fit beyond education and age indicators. Our findings underscore the dominant role of demographic structure and education over physical access in shaping vaccine coverage, emphasizing the importance of localised, equity-focused strategies for future public health interventions in Ireland.

Keywords COVID-19 · Vaccination Rate · Geospatial Analysis · Ireland · LEA level

1 Background and Introduction

Background

COVID-19 vaccination has been one of the most impactful public health tools in mitigating the global burden of the pandemic. According to estimates published in The Lancet Infectious Diseases, vaccines prevented approximately 20 million deaths in their first year of rollout, highlighting their role in reducing transmission, hospitalization, and mortality in high-coverage settings(Watson et al. 2022). However, significant disparities in vaccine uptake have persisted across regions and population groups, reflecting deeper socio-political and structural divides.

A growing body of evidence underscores the influence of socio-demographic factors on vaccine acceptance. A global systematic review by Sallam in 2021(Sallam 2021) found that individuals with higher levels of education, income, and more trust in government and health authorities were more likely to accept COVID-19 vaccines. In contrast, mistrust, misinformation, and perceived risks were disproportionately prevalent in marginalized and under-served populations. Mohammad S Razai introduced the "Five Cs" framework—Confidence, Complacency, Convenience, Calculation, and Collective responsibility (Razai et al. 2021)- as a useful model for understanding vaccine hesitancy at the individual level. Other studies(Troiano and Nardi 2021), found that lower vaccine willingness was more common among women, younger adults, and ethnic minorities.

Spatial Patterns: While much of the literature on vaccine hesitancy has focused on individual-level attitudes and beliefs, an emerging body of research highlights the critical role of structural barriers—particularly geographic access—in shaping vaccination uptake. Spatial and logistical constraints—including distance to vaccination centers, lack of public transport, or regional disparities in health infrastructure—have been cited as major obstacles, even in high-income countries.

Evidence from England: Research examining spatial patterns in COVID-19 vaccination uptake has provided valuable insights into geographic disparities. A notable study (Dropkin 2022)of COVID-19 booster uptake across England found substantial variation, with Third Injection uptake rates ranging from 27.2% in Newham to 67.4% in Gloucestershire—representing a 40 percentage point difference between areas. This research examined area-level predictors beyond typical individual characteristics such as age, gender, ethnicity, and deprivation levels. The findings revealed that while individual socio-demographic factors remained significant predictors of vaccination uptake, considerable unexplained area-level variation persisted, suggesting that geographic factors play an important role in vaccination decision-making processes.

Research employing geographically weighted regression (MGWR) models (Chen et al. 2023) investigated how relationships between socio-demographic variables and vaccination rates vary across different geographic locations. The study found that social, economic, and demographic determinants explained 83.2% of the variance in vaccination uptake patterns, with the strongest positive associations observed for proportion of population over 40 years, car ownership rates, and average household income. Importantly, this research demonstrated that the strength of associations between predictors and vaccination uptake varied significantly across geographic areas, indicating that traditional global statistical models may not adequately capture local variations in these relationships.

Complementing these behavioural insights, recent studies have begun to quantify spatial inequities in vaccination service availability. A UK-based analysis using Gini coefficients revealed that rural areas consistently experienced poorer geographic access to vaccine centers compared to urban counterparts (Chen et al. 2024).

Ireland's Vaccination Strategy: Ireland's COVID-19 vaccination campaign was supported by a diversified service delivery network designed to maximise accessibility, equity, and speed of distribution. This network was composed of three primary modes of vaccine administration: mass vaccination centres, general practitioners (GPs), and community pharmacies. The configuration and expansion of these services were shaped by evolving public health priorities, population needs, and logistical feasibility throughout the roll out period.

Mass Vaccination Centres: A total of 35 large-scale vaccination centres ("COVID-19 Vaccination Centres and Walk-Ins - HSE.ie" 2021) were established across the country to serve as the backbone of the national rollout strategy. These centres operated in sports arenas, conference halls, and other large public venues, enabling high-throughput administration to priority populations, especially during early phases of the campaign. Their geographic distribution was determined by regional population density and accessibility.

GPs: They played a pivotal role in reaching population groups that required continuity of care, personalisation, or proximity-based access. The national registry of GP clinics participating in the vaccination programme included 1,557 sites ("Find a GP in Ireland" n.d.), spanning both urban and rural regions. Their involvement was particularly crucial in serving older adults, individuals with chronic illnesses, and those less likely to attend mass vaccination centres.

Pharmacies: A total of 1,041 pharmacies ("Find a Vaccine Location in Ireland" n.d.) were authorised to administer COVID-19 vaccines as part of the community-based delivery model. Their inclusion marked a strategic expansion of the vaccination network, leveraging existing infrastructure to facilitate walk-in appointments, flexible scheduling, and enhanced accessibility—particularly for younger adults and working populations. Pharmacies became especially critical from June 2021 onward (Ireland 2022), when the 18–34 age group became eligible for vaccines outside of scheduled HSE pathways.

Barriers to Vaccination: Despite Ireland achieving over 90% uptake of at least one COVID-19 vaccine dose among adults by the end of 2021, as reported by the Health Protection Surveillance Centre (HPSC) (Health

Protection Surveillance Centre 2021a, 2021b), spatial inconsistencies emerged at finer geographic scales. Disaggregated data at the LEA level revealed notable disparities, particularly in communities marked by higher socio-economic deprivation and limited healthcare access. These sub-national variations in vaccine uptake disparities remain insufficiently explored in the Irish research landscape and have yet to be systematically modelled using integrated spatial, demographic, and temporal data sources.

In contrast, spatial epidemiological approaches have proven effective in other countries. For instance, de Figueiredo (2021) applied multilevel regression and post-stratification techniques to map vaccine hesitancy across the UK, identifying clusters of low intent that correlated strongly with ethnicity, economic precarity, and urban residency (Figueiredo 2022). Similarly, McGowan et al. (2022) demonstrated that even in high-income settings with wide vaccine availability, uptake remained uneven—often reflecting underlying inequalities tied to deprivation levels (McGowan and Bambra 2022) but such approaches remain limited in the Irish context and suggest the need for sub-national, spatial-based modelling approaches in Ireland.

Introduction

Although Ireland's national COVID-19 vaccination campaign achieved high overall coverage by the end of 2021, substantial variation exists beneath these national averages (Health Protection Surveillance Centre 2021b). Emerging evidence suggests that vaccine uptake has varied significantly across regions and population subgroups, shaped by underlying social, economic, and geographic inequalities. Understanding why uptake differs between communities is essential for designing resilient and equitable public health systems.

While global literature often emphasises individual-level attitudes or broad national trends, fewer studies examine how spatial and structural factors intersect at local levels to shape vaccination outcomes. In Ireland, LEA-level data offer a valuable but underutilized opportunity to explore these disparities more granularly. Existing reports from the CSO and the Health Protection Surveillance Centre (HPSC) provide descriptive LEA-level statistics but do not systematically model the determinants of uptake using integrated spatial, demographic, and temporal data.

This thesis addresses this gap by investigating the underexplored intersection of spatial access, demographic structure, and vaccine uptake in Ireland. Specifically, it examines how factors such as age composition, educational attainment, general health outcomes, gender, unemployment, and proximity to vaccination centers are associated with local COVID-19 vaccine coverage for individuals aged 12 and older. Using both exploratory and statistical techniques—including Bayesian beta regression, CAR models, and sigmoidal curve-fitting—this study models spatial and temporal variation in vaccine uptake across Ireland's 166 LEAs.

By adopting a spatial-temporal lens and integrating diverse public data sources, this thesis contributes to the growing field of place-based public health research. The findings offer empirical insights that can inform more geographically targeted, demographically sensitive, and equity-driven immunization strategies for future pandemic responses in Ireland and comparable settings. .

2 Methods

2.1 Study Design

This study employs a quantitative, observational, and ecological research design to explore variation in COVID-19 vaccination uptake across Ireland. The ecological framework is suited to the research aims, as the analysis is conducted at the population level rather than the individual level, with the LEA serving as the unit of observation. This design enables the investigation of associations between area-level characteristics—such as age structure, educational attainment, gender distribution, and deprivation levels—and LEA-level vaccination uptake rates.

Observational designs are appropriate when experimental manipulation is neither ethical nor feasible, as is the case with vaccine uptake during a pandemic. This study does not aim to infer causality but rather to identify statistical associations and spatial patterns that may inform future public health strategies. While the ecological approach limits the ability to capture individual-level behavioural determinants or utilize more granular spatial units, it offers valuable insights into structural and contextual influences affecting access and uptake at the community scale. Moreover, although individual-level or finer-resolution data could allow for deeper behavioural inferences, such data were not available for this study. Future research incorporating more granular or person-level data could complement these findings and help disentangle behavioural mechanisms from broader structural trends.

Additionally, the study incorporates a temporal dimension by analysing vaccination uptake over successive months, enabling the identification of evolving patterns and time-based disparities in vaccine coverage. This temporally extended approach complements the spatial analysis and enhances the explanatory power of the models. The combination of spatial and temporal design elements positions this research within the growing body of data-driven public health studies aimed at understanding complex, dynamic health phenomena within real-world settings.

2.2 Data and Variable Preparation

2.2.1 Data Sources and Description

This study draws on a range of publicly available datasets and administrative sources relevant to Ireland's COVID-19 vaccine rollout, population characteristics, and geographic boundaries. All data were aggregated at the LEA level to enable spatial analysis at a meaningful sub-national resolution.

Vaccination Infrastructure: Data on the locations of mass vaccination centres (35 in total), GP clinics (1557), and community pharmacies (1041 in total) were obtained from the official HSE website. These represented the full network of vaccine delivery sites active during the 2021 national roll out "Find a Vaccine Location in Ireland" (n.d.).

Vaccine uptake rates were sourced from the CSO website, specifically Table CDC47(n.d.a), which provides the proportion of the population vaccinated witn the primary vaccine dose, booster 1 and booster 2 at the LEA level every month from 2021 to 2023. This dataset is the most granular, consistent public source of COVID-19 vaccine coverage in Ireland.

Demographic indicators were obtained from the CSO's 2022 Census small-area population statistics and included the following:

Age: Age data was extracted from SAP2022T1T1LEA22(n.d.b), individual-level age data were to be subsequently aggregated into five analytically relevant age groups (12–17, 18–54, 55–64, 65–70, and 71+ years).These groups were not selected arbitrarily, but to reflect the sequenced eligibility criteria of Ireland's COVID-19 vaccination roll out, which was largely structured by age and risk)(Government of Ireland 2020).

The national campaign began on 29 December 2020 with the vaccination of frontline healthcare workers and residents in long-term residential care settings. This was followed by a sequential age-based expansion, beginning with adults aged 85 years and over, and progressing in five-year increments down to those aged 65 by March 2021. The roll out then extended to adults aged 50–64 by May 2021, and subsequently to individuals aged 35–49. The most significant expansion occurred in late June 2021, when vaccination was made available to all adults aged 18–34 years through community pharmacies "Responses Archive" (n.d.).

In August 2021, the programme was further extended to include adolescents aged 12–15 years, based on updated recommendations by the National Immunisation Advisory Committee (NIAC)("Minister for Health Confirms That Ireland's COVID-19 Vaccination Programme Will Open to 12-15 Year Olds" n.d.). However, children aged under 12 were not eligible for vaccination during the study period. Although vaccines were later approved for children aged 5–11 in early 2022, roll out began only in January of that year and was not accompanied by a comprehensive or sustained national campaign. Coverage in this group remained limited and inconsistent across LEAs and was a reason we did not focus on them in our study.

The details of the age categories:

- 12–17 years: This group became eligible for vaccination in August 2021, with eligibility requiring parental consent. They were primarily vaccinated through schools or dedicated youth clinics.
- 18–54 years: Representing the largest working-age population, this group accessed vaccines through mass vaccination centres, general practitioners, and pharmacies.
- 55–64 years: Considered higher-risk due to age-related vulnerability and comorbidities; this group was prioritised earlier in the programme.
- 65–70 years and 71+ years: These were among the first groups offered vaccines, including individuals in congregated residential settings and older adults with mobility limitations.

This age categorisation ensures alignment with the actual vaccine distribution strategy while allowing for meaningful demographic distinctions in modelling.

Self-Reported Health Status: From SAP2022T12T3LEA22(n.d.c), respondents classified their general health using a 5-point Likert scale: - Very Good - Good - Fair - Bad - Very Bad These categories capture perceived physical and mental health, which may influence vaccine-seeking behaviour. And the proportion of people in each of these categories was to be extracted at the LEA level.

Educational Attainment: This data was drawn from SAP2022T10T4LEA22(n.d.d), where the thirteen original response categories were to be grouped into the six analytically meaningful levels: no formal education, primary education, upper secondary, apprenticeship/trade, honours bachelor degree, and postgraduate qualifications. These reflect major stages in the Irish education system and allow analysis of educational gradients in vaccine uptake.

Deprivation Indicators: To capture the structural socioeconomic context underpinning vaccine uptake variation, this study incorporated multiple area-level deprivation metrics derived from the Pobal Haase-Pratschke (HP) Deprivation Index(Pobal 2022).The Pobal HP Index is Ireland's most widely adopted small-area deprivation framework, integrating multiple census-based dimensions—demographic profile, social class composition, and labour market status—into a single relative measure of affluence or disadvantage. All indicators were matched to the LEA level for spatial consistency.

Deprivation Score: This continuous variable serves as the core summary metric of area-level deprivation, derived via principal component analysis from multiple underlying census inputs. The score is centred around a national mean of zero, where positive values reflect relative affluence and negative values indicate increasing levels of disadvantage. This variable was critical for assessing socioeconomic gradients in vaccine uptake, particularly in the context of prior research suggesting lower vaccination rates in marginalised communities.

Unemployment Rates: Gender-disaggregated unemployment rates were included to capture labour market exclusion and economic insecurity. Male unemployment (UNEMPM22) and female unemployment (UNEMPF22) were treated as separate indicators given known gender disparities in employment trends and household responsibilities, both of which can influence access to healthcare services. These variables further provide insight into the economic stress profile of LEAs, which may affect not only vaccine uptake but also health-seeking behaviour more broadly.

Geographic Boundaries: LEA boundary shapefiles were obtained from Ordnance Survey Ireland (OSI) via the CSO GeoHive Portal("CSO Local Electoral Areas - National Statistical Boundaries - 2022 - Ungeneralised" n.d.). They provided the official digital geographic boundaries necessary for spatial analysis and mapping applications. These boundary files serve as the foundational geographic framework for linking vaccination rates with corresponding service locations.

2.2.2 Data Collection Methods

Each dataset underwent systematic extraction, cleaning, and integration using a combination of R, Python, and geospatial tools.

The following describes how data were collected and prepared for analysis.

Vaccination Centre Locations

Vaccination site data were obtained through web scraping techniques applied to the HSE website:

Static Web Pages: The rvest package in R was used for scraping data from static HTML pages. For static web pages, the information is directly available in the HTML source when the page loads, making rvest ideal for quick and simple extraction of data from the page's structure.

Dynamic Web Pages: Selenium in Python was utilized to handle JavaScript-rendered content, enabling access to dynamically loaded data not accessible via static scraping. Some websites use dynamic content, where information is loaded separately through JavaScript after the initial page is displayed. Selenium can simulate a real web browser, allowing the script to wait for content to load, click buttons, or scroll down the page to access hidden or delayed information. While rvest is faster and easier for static content, Selenium is necessary for interacting with and extracting data from more complex, dynamic sites.

All identified vaccination centers were geocoded to obtain their geographic coordinates (longitude and latitude). The Google Maps Geocoding API ("Google Maps Platform Documentation | Geocoding API" n.d.) was employed to ensure accurate and standardized geolocation across the dataset. A custom function was written to send HTTP requests to the Google API with each address and extract the corresponding latitude and longitude from the returned JSON data. This approach allowed precise control over the geocoding process and can be combined with rate limiting to comply with API usage policies.

Vaccination Uptake Data The CDC47 table from the CSO was downloaded in CSV format. Data were cleaned to remove duplicate records or inconsistent reporting dates. We restricted analysis to individuals aged 12 years and older, in line with our population of interest and matched vaccine uptake statistics to the correct LEA geometry.

Demographic and Deprivation Data

All demographic variables were accessed through CSO SAP datasets. Variables were to be recoded into grouped categories mentioned in section 3.1 to get marginal distributions. This would also be used to get the joint age-sex demographic proportions.

Deprivation and Unemployment Indicators were present in the 2022 Pobal HP Index was obtained by directly contacting a liaison working at Pobal and the data shared with us was downloaded in a CSV format.

2.2.3 Engineering Accessibility Data

Traditional accessibility measures include naïve distance-based metrics and provider-to-population ratios, Luo and Qi (Luo and Qi 2009)established the modern spatial accessibility analysis method of two-step floating catchment area (2SFCA). They demonstrated that the 2SFCA method addresses critical limitations of traditional distance-based measures by incorporating both healthcare service supply (provider capacity) and it's demand (population need) factors simultaneously. This approach provides more accurate representations of healthcare accessibility patterns by accounting for competition among populations for limited healthcare resources. Their method first involves identifying all population locations within a reasonable travel distance of each healthcare provider, then calculation of provider-to-population ratios. In the second step, it identifies all providers within reasonable distance of each population location and sums their accessibility ratios. This dual consideration of supply and demand makes the 2SFCA method relevant for our study where both vaccination site capacity and population demand vary spatially.

Luo and Wang(Luo and Qi 2009)provided a comprehensive comparison of gravity models, enhanced 2SFCA methods, and other spatial accessibility approaches. Through their analysis in the Chicago region, it was demonstrated that different accessibility measures can produce substantially different results depending on geographic context, population distribution, and healthcare service characteristics. The study revealed that gravity models, which weight healthcare opportunities by distance decay functions, often provide more nuanced accessibility scores than simple measures. Wang's framework offered crucial direction for vaccination accessibility research by establishing important factors to take into account when choosing suitable spatial accessibility models depending on study goals and data availability.

Nevertheless, detailed demographic statistics at the sub-LEA level and, ideally, information regarding service capacity are necessary for the complete implementation of 2SFCA, and these were not available for this study. Furthermore, the approach is resource-intensive, requiring substantial geospatial processing and catchment dynamics assumptions. So, we decided not to use 2SFCA in our analysis as a result.

We used an intuitive spatial coverage measure based on proximity - travel isochrones were generated for each service location. Specifically, 10, 20, 30, and 60-minute drive-time isochrones were created for the initial vaccination center, while 10-minute drive-time isochrones were generated for pharmacies and GPs. Each LEA polygon was then intersected with these isochrones to determine the extent of overlap, representing the proportion of the LEA covered by the service catchment areas. The total overlapping area was measured to produce a naive accessibility score for each LEA. For the initial vaccination center, a more refined accessibility score was calculated by inversely weighting the overlapping areas by travel time, reflecting the assumption that closer proximity implies better access. This approach was based on the intuitive premise that any spatial overlap between a healthcare service's catchment and an LEA indicates at least some degree of physical access for residents.

Accessibility Score Calculation:

$$C_{ijk} = \frac{I_{ijk}}{T_k}$$

$$A_{ij} = \sum_{k=1}^{K} C_{ijk}$$

$$A_{vacc} = \frac{1}{10} \cdot A_{i,10} + \frac{1}{20} \cdot A_{i,20} + \frac{1}{30} \cdot A_{i,30} + \frac{1}{60} \cdot A_{i,60}$$

$$A_{pharm} = A_{i,10}$$

Where C_{ijk} is the coverage share, $ I_{ijk}$ is intersection area, T_k is isochrone area, and A_{ij} is the accessibility score

Applying 2SFCA—or an advanced variation—would yield a more comprehensive assessment of healthcare accessibility with more specific data and specialized resources, particularly in contrasting rural and urban situations where demand and service burden varied significantly and we strongly advise using it in subsequent research.

2.2.4 Data Merging

A substantial data merging and transformation process was undertaken to compile the final analytical dataset used in this study. This dataset consisted exclusively of sociodemographic indicators, deprivation metrics, and accessibility scores—excluding any vaccination uptake data.

The vaccination uptake data with the LEA geometries remained separate and were merged with the final analytical dataset based on the research question like the vaccination dose of interest that was studied etc.

The age-related dataset originally included individual-level counts from age 0 to 19 and grouped intervals thereafter. The recoding involved aggregating the relevant age columns and reshaping the dataset using pivot_wider() and mutate() functions from the tidyverse package.

For health and education, only a subset of categories previously mentioned in 3.1 & 3.2 were retained to maintain analytical relevance and interpretability. Health status, initially reported in six categories, was reduced to five and Education was recoded from thirteen original categories into six.

Each of these demographic features was transformed into proportions to enable marginal distribution analysis at the LEA level. Additionally, joint distributions were derived to capture the proportion of males and females within each age category, allowing for sex-specific age structure assessments.

Among the many available variables from Pobal HP, only those of direct analytical relevance were retained: - Index22_rel: the composite deprivation index score - UNEMPM22: male unemployment rate - UNEMPF22: female unemployment rate

Accessibility data, calculated as discussed previously was also stored in a csv format mapped to LEA names.

To ensure consistency across datasets, LEA names were standardised using string processing functions such as str_replace_all() and str_to_title() to correct for spelling, punctuation, and formatting discrepancies. Merging was conducted using left_join() from the dplyr package, maintaining a one-row-per-LEA structure in the final output. The resulting dataset comprised 166 observations—one for each Local Electoral Area in Ireland—and included:

- Marginal and joint demographic distributions (age, sex, health, and education)
- Selected deprivation indicators (index score and gender-specific unemployment)
- 3 accessibility scores for the 3 different vaccination infrastructure

This merged dataset formed the statistical foundation for the exploratory and modelling phases of the study.

2.2.5 Model

We employed a hierarchical Bayesian modelling framework to examine determinants of primary COVID-19 vaccination uptake rate across LEAs. A total of six modelling approaches were implemented, each designed to address specific research questions, including bounded proportional outcomes, spatial autocorrelation, and nonlinear temporal trends. The models progressively build on one another, beginning with basic compositional demographic factors of age and gender and expanding to include compositional health and education metrics, spatial structure, deprivation, interaction effects of age and gender, and temporal dynamics.

All models were implemented using the brms package, a Bayesian regression modelling interface to Stan, which supports complex hierarchical, spatial, and temporal modelling structures, and leverages Markov Chain Monte Carlo (MCMC) sampling for full posterior inference "Parameterization of Response Distributions in Brms" (n.d.).

The primary outcome variable was the final rate of primary COVID-19 vaccination (bounded between 0 and 1), with booster dose data available for future extensions. The compositional nature of demographic covariates (e.g., age, sex, education, health) required Additive Log-Ratio (ALR) transformations to avoid issues of perfect multicollinearity due to unit-sum constraints. The baseline chosen for the transformation were the following: - Age: 71+ year olds - Education: Post Graduate Level - Health Outcome: Very Good - Sex: Female, For the joint distribution the baselines were 71+ year old males for the male group and 71+ year old females for the female group.

All cross-sectional models employed Beta regression with a logit link function, which is well-suited for modelling proportional outcomes constrained within the (0, 1) interval. In this framework, the mean of the Beta distribution—representing the expected value of vaccine uptake—is modelled as a function of the covariates via the logit link. While the Beta distribution allows for heteroskedasticity through its precision parameter (phi), in our models, phi was assumed to be constant across all observations and was not modelled as a function of predictor variables.

Model 1 focussed on basic demographic variables—namely age and gender—relate to the final uptake of the primary COVID-19 vaccine dose across LEAs. Bayesian inference was conducted via MCMC sampling with four chains of 2,000 iterations each (including 1,000 warm-up iterations), resulting in 4,000 post-warmup posterior draws and this was the functions' default in R.

Primary_Vax_Rate ~ age_12to17_logratio + age_18to54_logratio + age_55to64_logratio + age_65to70_logratio + male_logratio

The base model was extended to model 2 by adjusting for health, education, and accessibility to initial vaccination center. The accessibility measure was particularly relevant to the primary dose, which was more likely administered at centralized mass vaccination centres as opposed to decentralized locations like GPs or pharmacies, which were more important for booster doses. This model retained the beta regression framework and used the same MCMC setup as Model 1. Incorporating these additional variables allowed us to capture social determinants of health and logistical barriers to access, thereby enriching our understanding of factors influencing vaccine uptake beyond demographic composition alone.

Primary_Vax_Rate ~ age_12to17_logratio + age_18to54_logratio + age_55to64_logratio + age_65to70_logratio + edu_NoFormal_logratio + edu_Primary_logratio + edu_UpperSecondary_logratio + edu_Apprenticeship_logratio + edu_HonoursBachelor_logratio + health_VeryBad_logratio + health_Bad_logratio + health_Fair_logratio + health_Good_logratio + male_logratio + Wt_accessibility_Initial_Vacc

Model 3: Morans'I (Moraga n.d.) was calculated to identify if spatial autocorrelation existed in the residuals. Moran's I is a global measure of spatial autocorrelation that quantifies the degree to which similar values cluster together in geographic space. The presence of significant spatial clustering suggested that residuals were not independent across LEAs, violating a core assumption of standard regression models. To address this, we introduced a spatial modelling approach using CAR priors ("Spatial Conditional Autoregressive (CAR) Structures — Car" n.d.). This model built upon the covariates included in Model 2 and incorporated spatially structured random effects via an adjacency matrix representing geographical contiguity among LEAs. The CAR prior allows neighbouring areas to "borrow strength" from each other by sharing information, thus improving parameter estimation in areas with sparse or noisy data. Because spatial models are more complex and prone to convergence issues, we extended the MCMC sampling regime by trial and error and finally set it to 30,000 iterations per chain with 10,000 warm-up iterations, resulting in a total of 80,000 post-warmup draws. We also increased adapt_delta to 0.999 and set max_treedepth to 20 to stabilize sampling using the No-U-Turn Sampler (NUTS) in the same fashion. This spatially informed model was crucial for appropriately accounting for geographic dependencies in vaccination rates that may arise from shared infrastructure, regional policies, or cultural clustering.

Primary_Vax_Rate ~ age_12to17_logratio + age_18to54_logratio + age_55to64_logratio + age_65to70_logratio + edu_NoFormal_logratio + edu_Primary_logratio + edu_UpperSecondary_logratio + edu_Apprenticeship_logratio + edu_HonoursBachelor_logratio + health_VeryBad_logratio + health_Bad_logratio + health_Fair_logratio + health_Good_logratio + Wt_accessibility_Initial_Vacc + car(lea_mat, gr = lea_id)

To further capture the socioeconomic context beyond individual-level education indicators, the fourth model introduced a deprivation index, representing a composite measure of material and social disadvantage at the area level. This variable was added to the extended multivariable model (Model 2) and allowed us to quantify how deprivation might influence vaccine uptake independently of individual education. The deprivation index was not compositional in nature, so it was included in its raw form. This model retained the beta regression framework and standard MCMC settings. By disentangling the effects of both individual (e.g., education, health) and area-level (e.g., deprivation) determinants, this model aimed to improve explanatory power and helped identify if communities could potentially be vulnerable to lower vaccine uptake due to systemic disadvantage.

Primary_Vax_Rate ~ age_12to17_logratio + age_18to54_logratio + age_55to64_logratio + age_65to70_logratio + edu_NoFormal_logratio + edu_Primary_logratio + edu_UpperSecondary_logratio + edu_Apprenticeship_logratio + edu_HonoursBachelor_logratio + health_VeryBad_logratio + health_Bad_logratio + health_Fair_logratio + health_Good_logratio + male_logratio + Wt_accessibility_Initial_Vacc + Depriviation

With model 5 we departed from the additive structure of previous models and explicitly modelled the joint distribution of age and gender. This approach allowed us to identify group-specific patterns that could not be detected when age and sex were considered independently. For example, vaccine hesitancy might be disproportionately higher among younger males compared to older females. To capture such differential effects, we included ALR-transformed proportions of eight age-gender combinations (e.g., females aged 12–17, males aged 55–64, etc.), using females and males aged 71+ as the reference group. This model did not include additional covariates like education or health and focused solely on the interaction structure. Like earlier models, it employed beta regression with a standard MCMC sampling regime of 4,000 post-warmup draws. This model was aimed at offering a more granular understanding of population heterogeneity in vaccine uptake.

Primary_Vax_Rate ~ alr_age_12to17_females + alr_age_18to54_females + alr_age_55to64_females + alr_age_65to70_females + alr_age_12to17_males + alr_age_18to54_males + alr_age_55to64_males + alr_age_65to70_males

While the previous models were cross-sectional in nature, model 6 was used to introduce a longitudinal perspective to capture how vaccination rates evolved over time within each LEA. We observed that vaccination uptake followed a characteristic sigmoidal (S-shaped) growth curve—slow at first, then rapidly increasing before eventually plateauing. To model this trajectory, we specified a four-parameter logistic function within a hierarchical Bayesian framework. The function estimated key curve parameters: base (the lower asymptote or initial vaccination rate), Asym (the upper asymptote or finalsaturation level), xmid (the inflection point where uptake accelerates most rapidly), and scal (the spread or steepness of the curve). LEA-specific random intercepts were included for base, Asym, and xmid, while scal was assumed to be globally constant due to model identifiability constraints. Furthermore, we modeled Asym as a function of ALR-transformed covariates (e.g., age and gender composition) to understand how basic demographic characteristics shape the final uptake level.The pother parameters can also be modelled as a function of the covariates in the future.

Priors for the nonlinear parameters were carefully chosen based on empirical inspection of the data. Specifically, we set a normal(0.8, 0.1) prior on Asym, reflecting the assumption that final vaccine coverage would typically plateau around 80%, with moderate uncertainty. This aligns with observed saturation levels in many LEAs. For xmid, we used a normal(6, 2) prior, based on the understanding that the inflection point—the month with the fastest uptake—was likely to occur within the first half-year after vaccine roll out. This prior captures temporal heterogeneity while remaining centered around plausible empirical timing. Finally, we applied an exponential(1) prior (constrained to be positive via lb = 0) on scal, ensuring strictly positive steepness while favoring gradual curves unless strong data signals indicate rapid growth.

The model was fitted to 4,980 monthly observations across 166 LEAs using the No-U-Turn Sampler (NUTS) with four chains of 4,000 iterations (2,000 warm-up), yielding 8,000 post-warmup draws. This hierarchical model would allow us to disentangle local variations in the speed and extent of vaccine uptake over time, offering valuable insights into both structural and behavioral determinants of vaccination trajectories.

Primary_Vax_Rate ~ base + (Asym - base)/(1 + exp((xmid - Month_num)/scal

Each of the four parameters in the logistic function plays a distinct role in shaping the uptake curve. The base parameter determines the initial starting point of vaccination uptake; higher base values indicate earlier initial uptake, potentially reflecting early access among priority groups. The Asym parameter governs the upper plateau, representing the final saturation level of vaccine coverage within each LEA. The xmid parameter controls the timing of the inflection point—the point at which uptake accelerates most rapidly—and is therefore key to identifying when vaccination momentum peaked. Finally, the scal parameter influences the steepness or spread of the curve; smaller values produce steeper curves, indicating faster uptake transitions, while larger values suggest a more gradual increase. By varying each of these parameters independently, the model captures a wide range of realistic vaccine uptake trajectories, from early surges to delayed or flattened rollouts. This flexibility is critical for understanding the diverse temporal dynamics of vaccine adoption across heterogeneous regions.

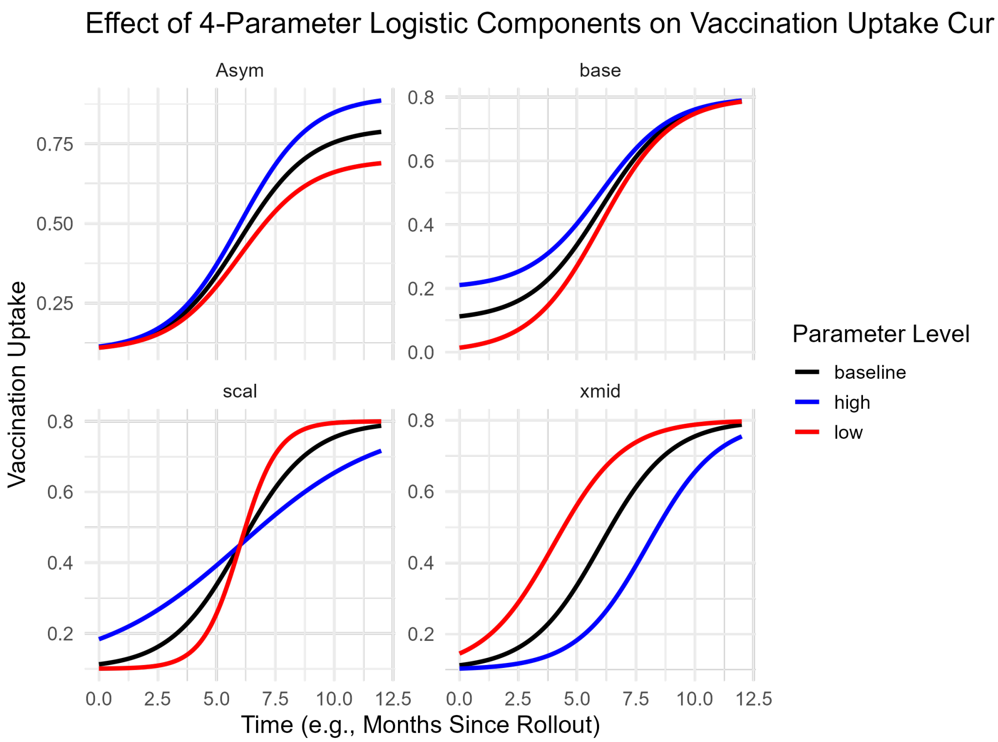

Figure 1: Effect of 4-parameter logistic model components (asymptote, base, scale, and midpoint) on vaccination uptake curves over time

3 Results

3.0.1 Descriptive Analysis

An initial exploration of the dataset was undertaken to evaluate the distribution and variability of key demographic and structural indicators across Ireland's LEAs. This section provides a descriptive overview of the demographics - age composition, health status, educational attainment, vaccination uptake, and deprivation levels across LEAs. These contextual factors are critical to understanding regional variability in COVID-19 vaccination patterns, particularly in relation to structural inequalities and accessibility barriers.

Age Composition: The 12–17 group represents adolescents, who required parental consent and were often vaccinated in dedicated clinics or school-associated pathways. The 18–54 bracket includes the bulk of the working-age population, who received vaccines through mass centres, general practitioners, and pharmacies. Individuals aged 55–64 were classified as higher risk due to age and comorbidities and were vaccinated earlier in the rollout. The 65–70 and 71+ groups were among the first prioritised by the state, especially those in congregated settings or with mobility and chronic health issues. These distinctions are critical not just from a clinical perspective, but also from a policy and accessibility standpoint, as they likely shape the speed, reach, and structure of vaccine deployment.

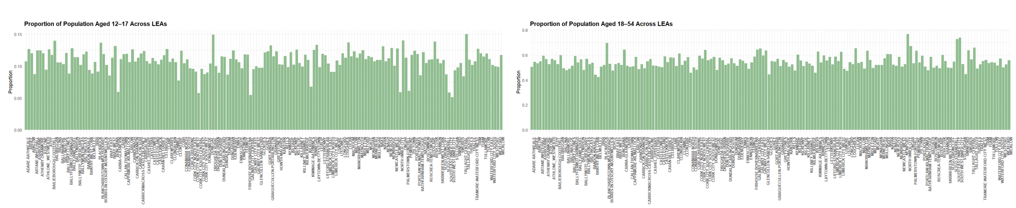

Figure 2: Age Proportion Distributions: 12–17 and 18-54

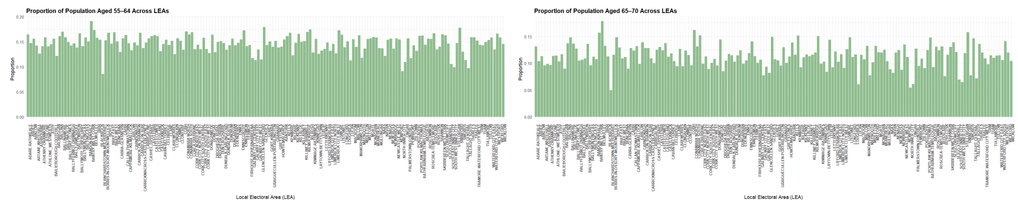

Figure 3: Age Proportion Distributions: 55-64 and 65-70

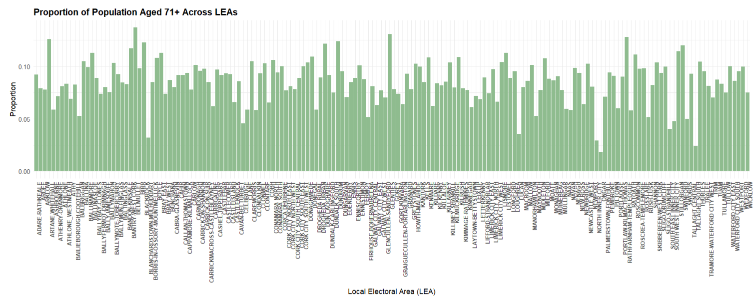

Figure 4: Age Proportion Distribution: 71+

The 18–54 group consistently constituted the largest segment of the population, with proportions ranging from approximately 42% to 77%, reflecting the economically active, working-age cohort. The 12-17 age group, comprising adolescents, represented between 5% and 15% of local populations, with slightly elevated proportions observed in suburban and commuter-belt LEAs. The 55–64 and 65–70 cohorts showed moderate variation, each accounting for up to 20% and 18% respectively in several rural areas, suggesting pockets of pre-retirement and early-retirement demographics. The 71+ age group displayed marked geographic variability, with certain western and peripheral LEAs reaching proportions exceeding 13%, aligning with broader demographic trends of ageing populations in rural Ireland.

Health Status: General health status is a core indicator of population well-being and serves as a proxy for the burden of chronic disease and overall quality of life. To visualize the geographical distribution of perceived health status across Ireland, proportions for each health category were calculated at the LEA level and presented using individual bar plots. These plots facilitate comparison across areas and allow for the identification of spatial clustering or outliers.

The proportion of residents reporting Very Bad health is consistently low across all LEAs, typically falling below 2%. However, subtle variations are observed, with urban-deprived LEAs such as North Inner City and Lifford-Stranorlar showing comparatively higher values. This spatial patterning corresponds with areas of higher socio-economic deprivation, suggesting a plausible link between living conditions and poor perceived health. In contrast, Bad health proportions are more variable, ranging between 3% and 8%. While still relatively low, this category demonstrates clearer urban-rural gradients. LEAs with higher unemployment and deprivation scores - particularly in the northwest and inner-city Dublin - tend to report elevated figures. The Fair health category captures a more neutral self-assessment and displays a broader distribution, with values typically between 15% and 30%. This group likely includes individuals managing chronic conditions without acute deterioration, and its prevalence further reflects socio-economic conditions, access to healthcare, and age structure of the population.

Good health reports constitute the modal response in most LEAs, suggesting a generally positive health self-perception across the country. The highest proportions are concentrated in suburban regions such as Ashbourne, Lucan, and Stillorgan, which may reflect younger population structures and better access to services.12–17 Finally, the Very Good category presents a more selective distribution, frequently ranging from 30% to over 50% in some eastern and commuter belt LEAs. The high concentration of Very Good responses in affluent LEAs likely reflects structural advantages such as employment stability, housing quality, and availability of preventive health services.

Education attainment: The spatial distribution of educational attainment across LEAs in Ireland was examined using the six distinct categories that were derived.

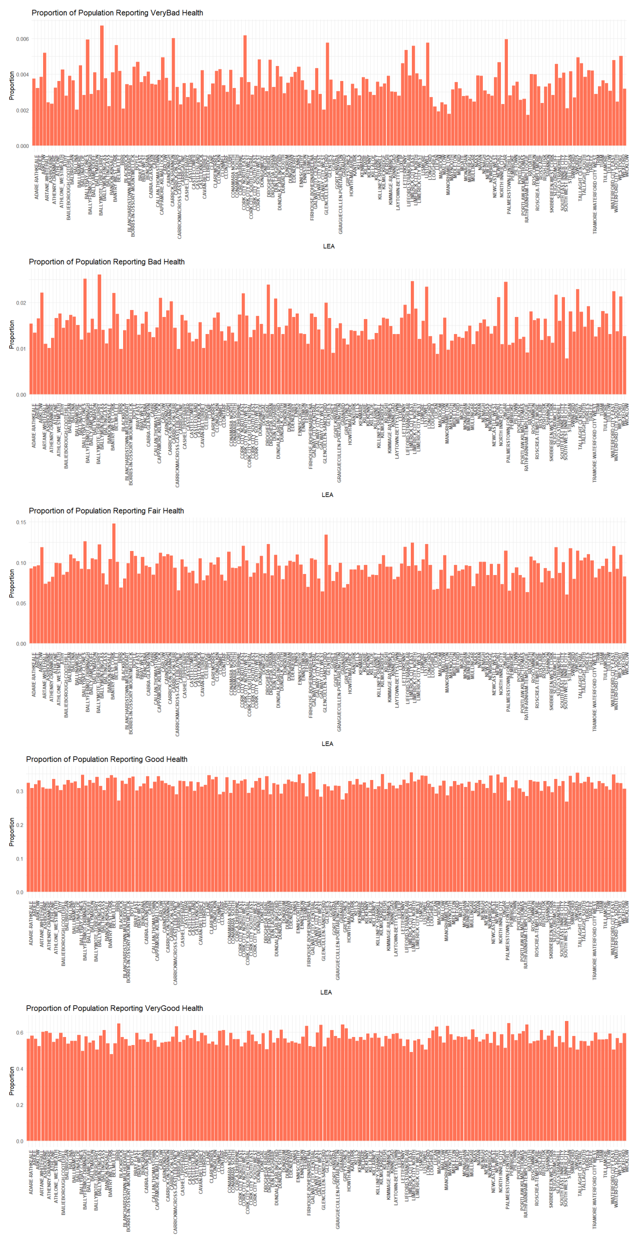

Figure 5: Proportion of self-reported general health status across LEAs

LEAs such as Dún Laoghaire, Blackrock, and Stillorgan consistently report higher proportions of individuals with honours bachelor and postgraduate degrees, highlighting the concentration of tertiary-educated populations in more affluent, urbanised areas. In contrast, rural and economically deprived regions-such as Donegal, Ballinamore, and Glenties-exhibit higher proportions of residents with no formal or only primary education.

Notably, Upper Secondary Education appears to be the most evenly distributed across LEAs, suggesting its widespread attainment across diverse socioeconomic settings. Meanwhile, the proportion of individuals completing apprenticeships varies substantially, with certain LEAs-especially those in semi-urban or industrial catchments-demonstrating a stronger presence of vocational education pathways.

Deprivation Scores: Across the 166 LEAs, deprivation scores ranged from approximately −21.6 (severe disadvantage) to +16.3 (marked affluence). LEAs such as Buncrana, Ballinamore, and North Inner City exhibited some of the lowest index scores, pointing to persistent socioeconomic deprivation. In contrast, LEAs like Blackrock, Dún Laoghaire, and Stillorgan consistently recorded positive scores, reflecting their concentration of economic advantage.

Sex-specific unemployment rates revealed complementary patterns. The male unemployment rate (UNEMPM22) ranged from under 3% to over 18%, with elevated levels in LEAs such as Donegal, Lifford-Stranorlar, and Monaghan. The female unemployment rate (UNEMPF22) also varied substantially, reaching as high as 22% in certain LEAs. While male and female unemployment rates often aligned geographically, some regions showed wider gender disparities.

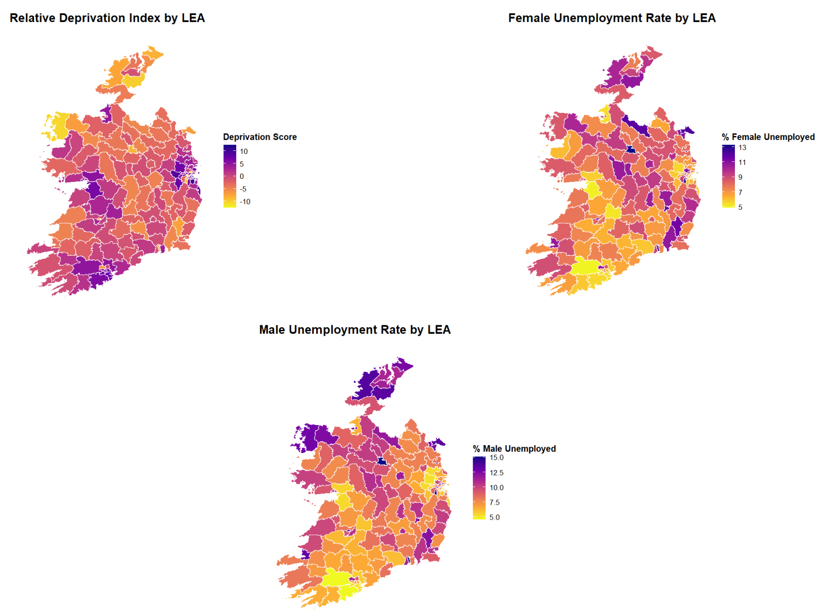

Figure 6: Distribution of deprivation metrics

3.0.2 Model

Model 1- Younger age groups demonstrated significantly lower vaccination rates compared to the reference category, with particularly strong negative effects for the 18-54 year age group and the 12-17 year age group.

The 55-64 year old variable was a strong positive predictor in the model. The credible interval (CI) for 65-70 year olds included zero, suggesting uncertainty around this effect. Male proportion demonstrated a negative association with vaccination rates but the CI included zero. The marginal effects plots are used to understand the impact to the predictors better.

Each curve's starting position on the x-axis represents the log-ratio of that age group relative to the elderly reference category (71+), illustrating the compositional distribution of age groups across Ireland's Local Electoral Areas (LEAs). For instance, the teal curve (ages 55–64) starts farthest to the right, around +1.8, indicating that in many LEAs this group is approximately six times more prevalent than the senior population (exp(1.8) approx 6). This pattern typically appears in workforce-heavy areas with a dominant working-age

Basic Demographic Model Results			
Variable	Estimate	95% CI Lower	95% CI Upper
(Intercept)	3.731	3.200	4.222
age_12to17_logratio	−0.607	−0.992	−0.226
age_18to54_logratio	−1.413	−1.755	−1.065
age_55to64_logratio	2.474	1.470	3.433
age_65to70_logratio	−0.421	−1.114	0.303
male_logratio	−1.782	−3.489	−0.057

Figure 7: Model 1 Summary Table

population. In contrast, the pink curve (ages 12–17) begins around –0.3, reflecting LEAs where adolescents are less numerous than seniors—often seen in aging or retirement-focused regions.

Because age proportions are compositional (summing to 1), increases in one age group imply decreases in others. This structural constraint is captured by the log-ratio approach, which allows meaningful comparisons while preserving the interdependence of demographic groups. The x-axis thus reflects relative demographic composition, while the y-axis shows the predicted outcome on the log odds scale—specifically, the log odds of the mean COVID-19 vaccination rate.

Importantly, LEAs with proportionally more adolescents relative to the elderly often exhibit lower vaccination rates. This is partly driven by the differing vaccine eligibility, uptake behavior, and hesitancy patterns across age groups. As one moves along the x-axis—reflecting underlying gradients such as area deprivation or socioeconomic variation—each curve shows how the log odds of vaccination change with demographic context. Points where the curves cross mark tipping zones where shifts in population structure coincide with shifts in vaccination behavior, offering insights into thresholds where targeted public health interventions may be most effective.

Model 2 – We could see that the age effects persist but attenuated slightly while revealing additional significant associations.

Upper secondary education emerged as a significant positive predictor, suggesting higher vaccination rates among populations with this educational level. Conversely, areas with higher proportions of individuals with no formal education showed lower vaccination rates. Health status variables showed no statistically significant associations with vaccination rates.

The male proportion effect changed to be positive but was still non-significant. Accessibility to initial vaccination centres showed non-significant impact suggesting that geographic accessibility was not a significant barrier to vaccination uptake in our population.

Model 3 - To formally test for spatial autocorrelation, we conducted a Moran's I test on the residuals: Moran's I = 0.1906, $p < 0.001$ — indicating significant positive spatial autocorrelation in residuals (i.e., LEAs geographically close to one another tend to have similar unexplained vaccination patterns) This confirmed that standard regression models failed to fully account for spatial dependencies, prompting the use of a CAR model, introducing a structured spatial random effect term that smooths values across neighboring areas The spatial correlation parameter (car) was estimated at 0.56 (95% CI: 0.03 to 0.99), suggesting moderate

Figure 8: Base Model Marginal Effects of Adjusted Model

Figure 9: Note: Log-ratios reflect the natural logarithm of the proportion of a group relative to the baseline group. This approach allows for comparison within compositional data.

positive spatial autocorrelation between neighboring LEAs, i.e., 56% of unexplained variance was attributable to spatial autocorrelation, indicating strong neighborhood-level similarities.

This spatial clustering may reflect unmeasured factors such as local social networks, community attitudes, healthcare provider practices, or other geographically-clustered influences on vaccination behavior. The relatively modest spatial standard deviation (0.21) indicates that while spatial correlation exists, the magnitude of unexplained spatial variation is limited, suggesting that the measured covariates capture much of the systematic variation in vaccination rates. Demographic patterns remained robust: Age 55–64 still had the strongest positive effect on uptake. Upper secondary education, remains a large positive driver and healthcare proximity and healthcare outcomes continued to show near-zero effects. The consistency of these effects across spatial and non-spatial models suggests that educational disparities in vaccination uptake operate independently of geographic clustering and supply-side geographic barriers and self-reported health outcomes were not significant determinants of vaccination disparities.

The inclusion of the Pobal HP Deprivation Index in *Model 4*, which measures relative affluence and disadvantage across Irish communities using composite scores derived from employment, education, housing quality, and demographic variables, had minimal impact on vaccination rates despite capturing broader socioeconomic conditions at the LEA level. We have seen that the deprivation scores ranged from approximately −21.6 (severe disadvantage) to +16.3 (marked affluence), representing a wide spectrum of socioeconomic conditions across all LEAs.

Extended Model with Education, Health & Accessibility			
Variable	Estimate	95% CI Lower	95% CI Upper
(Intercept)	3.574	0.844	6.205
age_12to17_logratio	−0.776	−1.398	−0.152
age_18to54_logratio	−1.419	−1.995	−0.827
age_55to64_logratio	1.954	1.001	2.885
age_65to70_logratio	0.268	−0.457	0.985
edu_NoFormal_logratio	−0.462	−0.826	−0.112
edu_Primary_logratio	−0.316	−0.805	0.151
edu_UpperSecondary_logratio	1.343	0.563	2.115
edu_Apprenticeship_logratio	−0.466	−1.127	0.215
edu_HonoursBachelor_logratio	−0.056	−1.183	1.085
health_VeryBad_logratio	0.402	−0.059	0.842
health_Bad_logratio	0.235	−0.562	0.997
health_Fair_logratio	−0.666	−2.148	0.857
health_Good_logratio	−0.866	−2.671	0.831
male_logratio	1.367	−0.645	3.383
Wt_accessibility_Initial_Vacc	−0.032	−0.092	0.028

Figure 10: Model 2 Summary Table

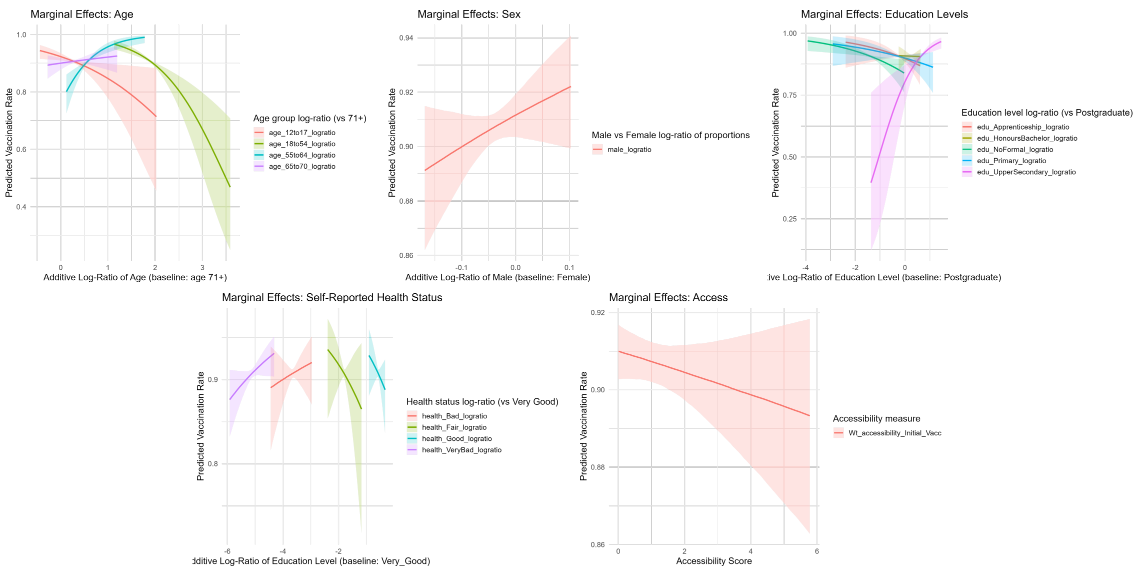

Figure 11: Marginal Effects of Adjusted Model

Figure 12: Note: Log-ratios reflect the natural logarithm of the proportion of a group relative to the baseline group. This approach allows for comparison within compositional data.

Despite this pronounced variation in area-level deprivation, the composite deprivation index showed a negligible association with vaccination rates, with confidence intervals including zero, suggesting that area-level socioeconomic deprivation does not provide additional explanatory power beyond the specific educational attainment variables already included in the model. This finding is particularly noteworthy given the substantial socioeconomic disparities observed across LEAs, indicating that educational composition effectively captures the most relevant aspects of socioeconomic variation affecting vaccination behavior. The results suggest that deprivation effects operate primarily through already-measured demographic and educational pathways rather than through additional unmeasured mechanisms. The lack of significant association indicates that while the Pobal index comprehensively measures community-level disadvantage across a wide range of conditions, individual educational characteristics remain the dominant socioeconomic predictor of vaccination uptake, effectively mediating the influence of broader area-level deprivation measures.

The age-sex interaction model *(Model 5)* revealed striking differences in age-vaccination relationships between males and females. Among females, younger age groups showed strong negative associations, with the 18-54 year group showing significant impact and among men the 65-70 year group uniquely showed a significant negative association contrasting sharply with the positive trend observed in older females, despite generally being considered a high-priority group.These results suggest that vaccine hesitancy patterns differ substantially between genders within the same age cohorts.

We found distinct patterns in COVID-19 vaccination uptake trajectories across Ireland's LEAs using our fitted hierarchical beta regression model *(Model 6)*. The population-level upper asymptote was at 0.94 (95% CI: 0.76-1.12) on the logit scale, representing the baseline maximum vaccination rate before accounting for demographic influences. The temporal inflection point occurred at 5.61 months (95% CI: 5.57-5.66), indicating that the steepest increase in vaccination rates typically occurred during late May to early June 2021, corresponding to the period when vaccine availability expanded beyond priority groups. The scale parameter was found to be 1.36 (95% CI: 1.35-1.38), which controls the steepness of the sigmoidal curve transition and suggests a relatively gradual uptake pattern.

We identified substantial between-LEA heterogeneity in vaccination achievements, with the asymptote random effect standard deviation of 0.81 (95% CI: 0.73-0.91) indicating considerable variation in maximum vaccination rates between areas. The base asymptote showed more modest variation with a standard deviation of 0.25 (95% CI: 0.22-0.29), while the timing of peak uptake demonstrated relatively low variation between LEAs and the inflection point random effect standard deviation was only 0.26 (95% CI: 0.23-0.30), suggesting moderately synchronized temporal patterns across LEAs despite substantial differences in final vaccination levels.

Spatial CAR Model Results			
Variable	Estimate	95% CI Lower	95% CI Upper
(Intercept)	2.890	0.465	5.404
age_12to17_logratio	−0.956	−1.567	−0.306
age_18to54_logratio	−1.405	−2.006	−0.816
age_55to64_logratio	2.293	1.307	3.260
age_65to70_logratio	0.219	−0.511	0.929
edu_NoFormal_logratio	−0.460	−0.828	−0.101
edu_Primary_logratio	−0.206	−0.675	0.269
edu_UpperSecondary_logratio	1.301	0.495	2.036
edu_Apprenticeship_logratio	−0.496	−1.148	0.197
edu_HonoursBachelor_logratio	0.019	−1.038	1.179
health_VeryBad_logratio	0.338	−0.105	0.781
health_Bad_logratio	0.265	−0.536	1.051
health_Fair_logratio	−0.913	−2.401	0.576
health_Good_logratio	−0.594	−2.318	1.137
Wt_accessibility_Initial_Vacc	−0.025	−0.084	0.035

Figure 13: Spatial Model Summary Table

The demographic predictors revealed significant positive associations between all covariates and final vaccination rate achievement. We found that more 65-70 years olds with respect to the 71+ year olds exhibited a strongest positive effect among age groups. It was also identified that LEAs with higher male proportions achieved substantially higher maximum vaccination coverage. The lower asymptote at -3.44 (95% CI: -3.49 to -3.39) on the logit scale, represents the minimal initial vaccination rates at the program's outset. Our demographic model maintained temporal stability, with the inflection point remaining consistent and the scale parameter showing minimal variation.

4 Discussions

The adjusted base model demonstrated superior fit, as evidenced by the increased precision parameter (phi = 77.74 vs. 57.18), indicating reduced unexplained variance. The persistence of age effects across both models underscores the robustness of demographic patterns in vaccination behaviour, while the emergence of educational effects highlights the importance of socioeconomic factors. The negligible impact of initial vaccination center accessibility suggests that supply-side constraints were not primary drivers of vaccination disparities in this context, pointing instead to demand-side factors related to demographic and socioeconomic characteristics.

Extended Model with Deprivation Index

Variable	Estimate	95% CI Lower	95% CI Upper
(Intercept)	3.836	0.985	6.594
age_12to17_logratio	−0.750	−1.382	−0.114
age_18to54_logratio	−1.422	−2.012	−0.830
age_55to64_logratio	1.903	0.957	2.839
age_65to70_logratio	0.282	−0.405	1.002
edu_NoFormal_logratio	−0.441	−0.821	−0.059
edu_Primary_logratio	−0.289	−0.781	0.208
edu_UpperSecondary_logratio	1.385	0.555	2.191
edu_Apprenticeship_logratio	−0.485	−1.192	0.219
edu_HonoursBachelor_logratio	−0.111	−1.279	1.100
health_VeryBad_logratio	0.405	−0.024	0.830
health_Bad_logratio	0.283	−0.526	1.060
health_Fair_logratio	−0.620	−2.072	0.881
health_Good_logratio	−0.963	−2.769	0.817
male_logratio	1.237	−0.967	3.308
Wt_accessibility_Initial_Vacc	−0.033	−0.093	0.028
Depriviation	0.010	−0.026	0.049

Figure 14: Model 4 Summary Table

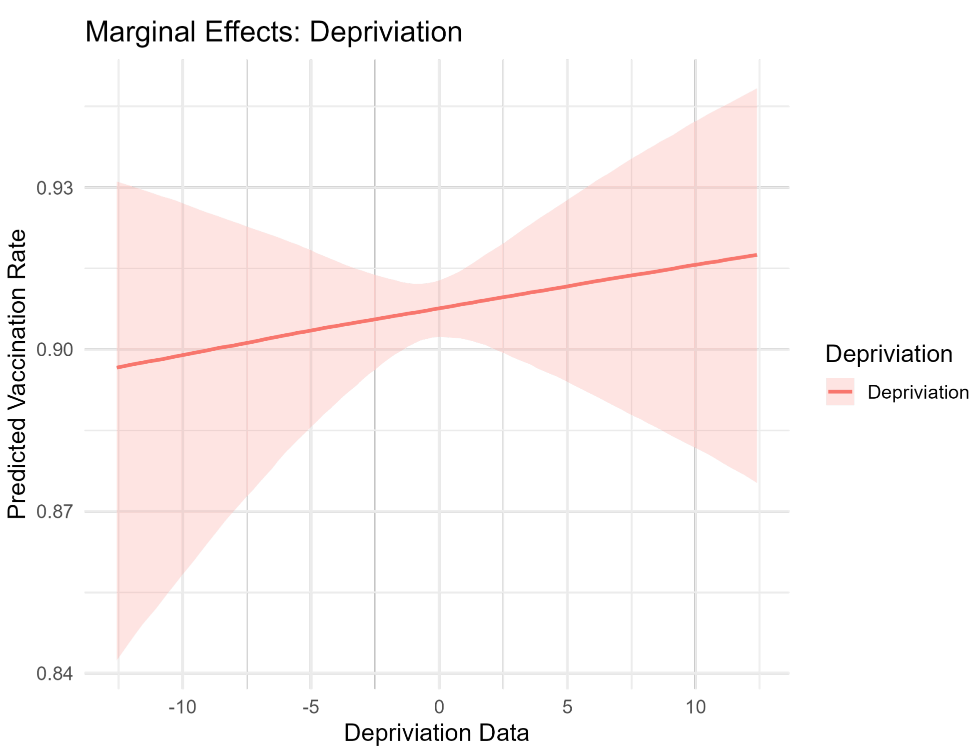

Figure 15: Marginal Effect of Depriviation

Age-Gender Stratified Model (ALR Transformed)

Variable	Estimate	95% CI Lower	95% CI Upper
(Intercept)	4.268	3.567	4.911
alr_age_12to17_females	−0.407	−1.610	0.797
alr_age_18to54_females	−2.169	−4.107	−0.259
alr_age_55to64_females	1.510	−0.170	3.230
alr_age_65to70_females	1.237	−0.287	2.761
alr_age_12to17_males	−0.097	−1.161	1.005
alr_age_18to54_males	0.402	−1.234	2.084
alr_age_55to64_males	1.430	−0.214	3.031
alr_age_65to70_males	−1.859	−3.338	−0.341

Figure 16: Model 5 Summary Table

The progressive increase in precision parameters across models (57.18 → 77.74 → 87.88) demonstrates continued improvement in model fit with the addition of socioeconomic variables and spatial structure and the spatial model represents the most comprehensive specification, accounting for both measured covariates and unmeasured spatially-correlated factors. The stability of key demographic and educational effects across all model specifications provides strong evidence for their robustness, while the significant spatial correlation highlights the importance of geographic context in understanding vaccination behaviour patterns.

We acknowledge that holding the precision parameter constant in the cross sectional models assumes homogeneous variance across LEAs. This may overlook meaningful differences in the uncertainty of vaccine uptake across areas. Future work could explore covariate-dependent precision modelling to better capture heteroskedasticity, particularly in relation to socioeconomic or spatial factors.

The longitudinal nonlinear model revealed nuanced demographic dynamics in vaccination uptake that differ from patterns observed in cross-sectional analyses. Specifically, the Asym was significantly and positively associated with the relative proportions of males and all working-age and older adult subgroups (ages 18–54, 55–64, and 65–70), with the strongest effects observed for the older cohorts. These findings suggest that, although younger adults and males may have exhibited delayed uptake in earlier phases of the campaign—as suggested in prior models—they ultimately reached high levels of vaccination over time. The estimated population-level midpoint for uptake (xmid aprox 5.61 months) indicates broad temporal alignment across LEAs, reflecting a relatively coordinated national rollout. However, substantial variation in asymptotic values between LEAs (sd(Asym) = 0.81) underscores the importance of local demographic composition in shaping final uptake levels. That the scalar parameter governing growth rate (scal = 1.36) was fixed across areas reinforces the interpretation that differences in eventual uptake—rather than uptake speed—were more influenced by population structure than by supply-side constraints. Together, these results indicate that extended temporal observation is critical for capturing the full extent of vaccination behaviour: initial disparities in uptake timing do not necessarily translate into persistent inequalities in ultimate coverage.

Based on the observed vaccination rate data, the increase-then-decline patterns of vaccination rate identified in LEAs such as South East Inner City, Maynooth, Galway City Central, Tramore-Waterford City West, and Kenmare suggest anomalies when compared to the typical sigmoidal vaccination uptake curve. A likely explanation for this pattern lies in changes to the underlying population denominators used in rate calculations. Since vaccination rates are computed relative to estimated local populations, any mid-series updates—such as those informed by the 2022 Census, student migration, or seasonal population shifts—could result in apparent declines despite stable or increasing absolute vaccination numbers. This effect is especially plausible in areas

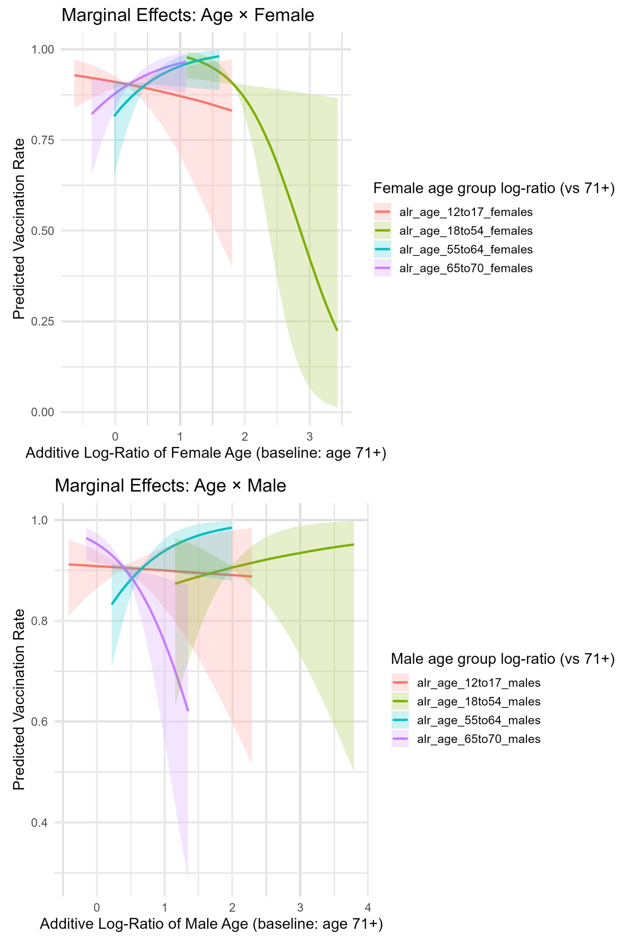

Figure 17: Marginal Effects of Age & Sex Interaction Model

Figure 18: Note: Log-ratios reflect the natural logarithm of the proportion of a group relative to the baseline group. This approach allows for comparison within compositional data.

with transient populations, such as university towns (Maynooth, Galway), city centres with high mobility (South East Inner City), and tourism-influenced or rural areas (Kenmare, Skibbereen-West Cork), where even small demographic changes can significantly distort percentage-based trends. As such, the observed dips are more likely to reflect population re-estimation or mobility rather than genuine decreases in vaccine uptake.

Our analysis relied on vaccination data reported only at the LEA level, not broken down by demographic subgroups or small area level data which was our biggest limitation. Population demographic characteristics were based on the 2022 Census and may not reflect the pandemic-period or vaccination roll out period scenario. Self-reported health status, used as a covariate, may be subject to bias. Our accessibility measure did not account for population demand or healthcare service's capacity, limiting its interpretive power; while a 2SFCA approach could address these spatial limitations, it requires more granular data and computational resources beyond this study's scope. Additionally, the analysis was restricted to the Republic of Ireland; cross-border vaccination—particularly in border regions—could have influenced local rates but remains unaccounted for in our models.

4.1 Code and Dashboard Availability

The code used in this thesis is available on GitHub:

GitHub Repository

An interactive Shiny dashboard showcasing accessibility measures:

Age-Gender Stratified Model (ALR Transformed)			
Variable	Estimate	95% CI Lower	95% CI Upper
(Intercept)	4.268	3.567	4.911
alr_age_12to17_females	−0.407	−1.610	0.797
alr_age_18to54_females	−2.169	−4.107	−0.259
alr_age_55to64_females	1.510	−0.170	3.230
alr_age_65to70_females	1.237	−0.287	2.761
alr_age_12to17_males	−0.097	−1.161	1.005
alr_age_18to54_males	0.402	−1.234	2.084
alr_age_55to64_males	1.430	−0.214	3.031
alr_age_65to70_males	−1.859	−3.338	−0.341

Figure 19: Temporal Model Summary Table

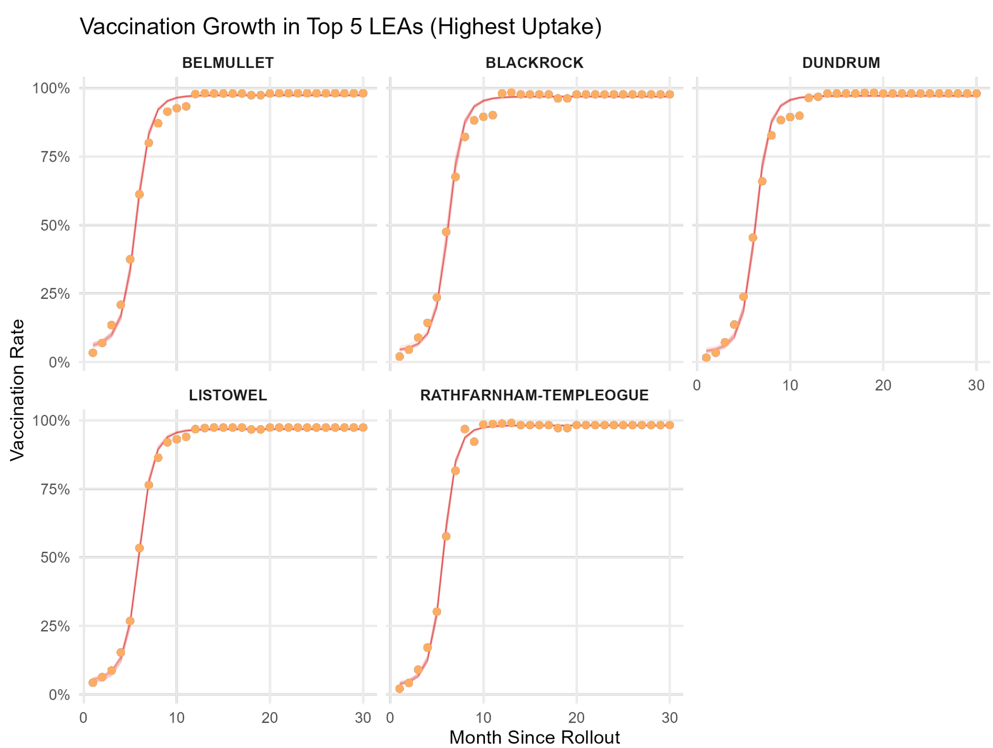

Figure 20: Vaccination Trend for Top 5 LEAs

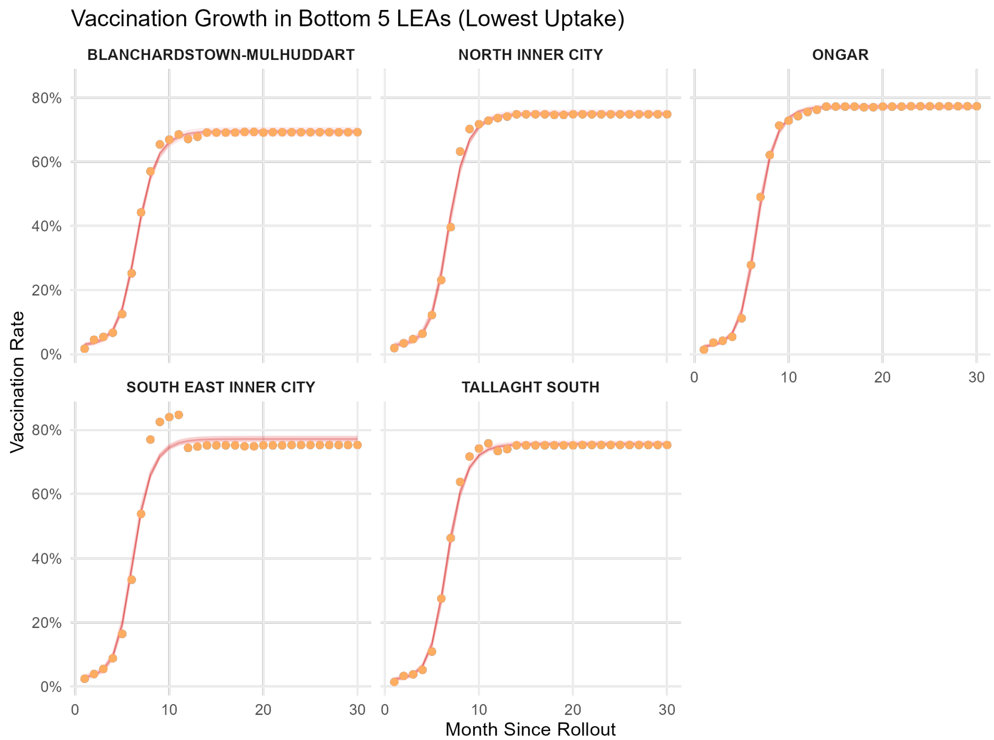

Figure 21: Vaccination Trend for Bottom 5 LEAs

Shiny Dashboard

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